

The Cosmic Yoyo: A Stick-Slip Brane Motor

Resolving Three Cosmological Anomalies via a Dynamical Attractor
on an ER=EPR Entangled Membrane with Israel Junction Conditions

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Abstract. The Λ CDM concordance model faces three robust observational tensions: DESI's 4σ detection of evolving dark energy, the persistent S_8 discrepancy between CMB and weak lensing, and Planck's unexplained low- ℓ ISW power deficit. We model the universe as a 3-brane oscillating in an AdS_5 bulk via a *non-linear stick-slip relaxation motor*. The radion field ϕ obeys $\ddot{\phi} + 3H\dot{\phi} + \xi R\phi + \partial_\phi V_{\text{GW}} = \mathcal{F}[E_{\mu\nu}] - \mathcal{R}\Theta(|\phi| - \phi_{\text{crit}})$, where the forcing $\mathcal{F}[E_{\mu\nu}]$ arises rigorously from the projected Weyl tensor $E_{\mu\nu}$ of the 5D bulk via Israel junction conditions (Shiromizu, Maeda & Sasaki 2000), and a non-minimal coupling $\xi R\phi$ ensures convergence to a *dynamical attractor* that locks the period at $T = 2.0$ Gyr despite evolving $H(t)$. BBN is protected by conformal symmetry: the radion couples to the trace $T^\mu_\mu = -\rho + 3p$ of the cosmic fluid, which vanishes rigorously during the radiation era ($w = 1/3$) and activates only after the QCD phase transition ($\tau_0^{1/3} = 257 \text{ MeV} \approx \Lambda_{\text{QCD}}$) breaks chiral symmetry. Topological anchoring is provided by primordial micro-PBHs with an extended mass function (log-normal, 10^{-14} – $10^{-10} M_\odot$), clustered near the brane and evading microlensing constraints via finite-source effects. The S_8 tension is resolved through scale-dependent Yukawa screening: $G_{\text{eff}}(k) = G_N(1 + \alpha e^{-k/k_L})$, suppressing growth at non-linear scales (DES/KiDS) while preserving standard gravity at linear scales (CMB). Phase coherence is ensured by ER=EPR entanglement [5]. The model resolves three established anomalies: (i) DESI's phantom crossing, (ii) S_8 tension via scale-dependent $G_{\text{eff}}(k)$, and (iii) Planck's ISW low- ℓ deficit ($\Delta\chi^2 = 32.9$). The definitive future test is SKA's detection of a 2 Gyr spatial modulation in the 21 cm power spectrum during the Epoch of Reionization. Laboratory falsification at $L = 0.2 \mu\text{m}$ is accessible via ultra-cold quantum neutron experiments (qBOUNCE at ILL) and levitated nanoscale optomechanics, which bypass the Casimir background that blinds macroscopic experiments at sub-micron scales.

1 Introduction

The concordance model faces three robust, independent anomalies. DESI's 2024–2026 data provide 4σ evidence for evolving dark energy [1,2]. The S_8 tension persists at $> 3\sigma$ between CMB and weak lensing surveys [3]. Planck observes an unexplained low- ℓ ISW power deficit.

We propose the universe is a 3-brane in a 5D Anti-de Sitter bulk, driven by a *stick-slip relaxation motor* whose forcing derives from the projected bulk Weyl tensor $E_{\mu\nu}$ via Israel junction conditions [13,14]. The bulk is a non-local topological state where spacetime is emergent [9]. Black holes form an ER=EPR entanglement network [5], ensuring global phase coherence. The brane is anchored by primordial

micro-PBHs with an extended mass function (10^{-14} – $10^{-10} M_\odot$), acting as topological capillaries [12].

The oscillation period $T = 2.0$ Gyr is locked by a dynamical attractor (non-minimal coupling $\xi R\phi$), calibrated from DESI BAO modulation and the ISW resonance at $\ell = 10$ –20 [6].

2 Stick-Slip Membrane Dynamics

The 5D action reads:

$$S = \int d^5x \sqrt{-g_5} \left(\frac{M_5^3}{2} R_5 - \Lambda_5 \right) + \int d^4x \sqrt{-g_4} (\mathcal{L}_{\text{br}} - \tau) \quad (1)$$

The brane position (radion field ϕ) obeys the

V7.1 *stick-slip motor* ODE with non-minimal gravitational coupling and trace-modulated forcing:

$$\ddot{\phi} + (3H + \Gamma_{\text{rad}})\dot{\phi} + \xi R\phi + \frac{\partial V_{\text{GW}}}{\partial \phi} = \mathcal{F}[E_{\mu\nu}] (1 - 3w) \mathcal{R} \Theta(|\phi| - \phi_{\text{crit}}) \quad (2)$$

Each term plays a distinct physical role:

- $3H\dot{\phi}$: Hubble friction (cosmological damping)
- $\xi R\phi$: Non-minimal coupling to the 4D Ricci scalar $R = 6(\dot{H} + 2H^2)$, ensuring convergence to a dynamical attractor that locks $T = 2.0$ Gyr despite evolving $H(t)$ and decaying accretion rates
- $\partial_\phi V_{\text{GW}}$: Goldberger-Wise restoring potential [11], with minimum at the QCD scale ($\tau_0^{1/3} = 257 \text{ MeV} = \Lambda_{\text{QCD}}$)
- $\mathcal{F}[E_{\mu\nu}]$: Geometric forcing from the projected bulk Weyl tensor. Via the Shiromizu-Maeda-Sasaki formalism [13], the Israel junction conditions relate the 5D Weyl tensor $C_{ABCD}^{(5)}$ to the effective energy-momentum on the brane. The electric part $E_{\mu\nu} = C_{AMBN}^{(5)} n^A n^B$ (where n^A is the brane normal) acts as a tidal force driving ϕ toward ϕ_{crit} . When micro-PBHs accrete dark matter, the local bulk curvature increases, amplifying $E_{\mu\nu}$ and thus the geometric forcing — a rigorous 5D gravitational mechanism requiring no information-theoretic intermediary
- $\mathcal{R} \Theta(|\phi| - \phi_{\text{crit}})$: Non-linear release activated by the Heaviside function Θ when $|\phi|$ exceeds the critical threshold ϕ_{crit} , triggering rapid energy discharge (the “slip”)

Dynamical attractor and period stability: A naive stick-slip motor would “chirp” as $H(t)$ decreases with cosmic expansion and DM accretion rates decay ($\propto a^{-3}$). The non-minimal coupling $\xi R\phi$ resolves this: the coupled system $\{H(t), \phi(t), \dot{M}_{\text{DM}}(t)\}$ converges to an attractor manifold where the competing effects of decreasing friction, decreasing forcing, and curvature feedback lock the period at $T = 2.0$ Gyr. Numerical integration confirms

convergence within ~ 2 e-foldings (Section 5). The oscillation cycle proceeds as: (1) *Stick phase*: $E_{\mu\nu}$ geometric forcing slowly charges ϕ toward ϕ_{crit} against the GW restoring potential $\partial_\phi V_{\text{GW}}$; (2) *Slip phase*: threshold crossing triggers rapid energy release, ϕ snaps back to equilibrium.

BBN protection via conformal symmetry: In braneworld effective actions, the radion couples to the trace of the energy-momentum tensor $T_\mu^\mu = -\rho + 3p$. The geometric forcing acquires a trace-coupling factor:

$$\mathcal{F}_{\text{eff}} = \mathcal{F}[E_{\mu\nu}] \times (1 - 3w_{\text{eff}}) \quad (3)$$

During the radiation era ($w_{\text{eff}} = 1/3$), conformal symmetry ensures $T_\mu^\mu = 0$ *rigorously*. The forcing vanishes identically, and extreme Hubble friction ($3H\dot{\phi}$) overdamps the radion. BBN proceeds with pristine $G_{\text{eff}} = G_N$. At the QCD phase transition ($T \approx 150 \text{ MeV}$), chiral symmetry breaking makes matter non-relativistic ($w \rightarrow 0$), the trace becomes non-zero ($T_\mu^\mu \approx -\rho$), and the coupling factor jumps from 0 to 1—igniting the stick-slip motor at exactly the QCD scale.

The resulting dark energy equation of state:

$$w(z) = -1 + A_w \sin\left(\frac{2\pi t_{\text{lb}}(z)}{T} + \phi_0\right) \quad (4)$$

with $A_w = 0.003 \pm 0.0005$ and $\phi_0 = \pi/2$, placing us at a *maximum* of $w(z) \approx -0.997$ today. Note: the stick-slip waveform is not purely sinusoidal (slower ramp, faster release), but Eq. 3 captures the leading harmonic.

Key parameters: $\tau_0 = 7.0 \times 10^{19} \text{ J/m}^2 = 0.017 \text{ GeV}^3$, $L = 0.2 \mu\text{m}$, $f_{\text{osc}} = 0.10$, $a_0 = 1.1 \times 10^{-10} \text{ m/s}^2$.

QCD connection: $E_\tau = \tau_0^{1/3} = 257 \text{ MeV} \approx \Lambda_{\text{QCD}}$. The brane tension is set by the strong force vacuum energy. This is not a free parameter — it emerges from the QCD confinement scale.

Adiabatic shield: The oscillation frequency ($\nu \sim 10^{-17} \text{ Hz}$) is 10^{31} times slower than the lightest KK mode ($\nu_{\text{KK}} \sim 10^{14} \text{ Hz}$). Branon production is suppressed by a Schwinger factor $\Gamma \propto e^{-10^{31}} \approx 0$.

Micro-PBH anchors (Extended Mass Function): The PBH population follows a log-

normal distribution [12,15]:

$$\frac{dn}{d \ln M} = \frac{n_0}{\sqrt{2\pi}\sigma_M} \exp\left(-\frac{(\ln M - \ln M_c)^2}{2\sigma_M^2}\right) \quad (5)$$

with $M_c \sim 10^{-12} M_\odot$ and $\sigma_M \approx 1.5$, spanning 10^{-14} – $10^{-10} M_\odot$. For this mass range, $r_s = 2GM/c^2 \approx 0.03$ – $300 \text{ nm} \sim \mathcal{O}(L)$. The extended mass function evades Subaru-HSC microlensing constraints [15] via finite-source effects, and PBH clustering near the brane further reduces their effective cross-section. These micro-PBHs ($\sim 10\%$ of dark matter) are the topological capillaries connecting brane to

bulk. JWST’s “Little Red Dots” are irrelevant — many are stellar clusters [10].

3 Resolving Three Anomalies

3.1 A. Evolving Dark Energy (DESI)

DESI favors dynamical dark energy at 4.2σ [2], with $w_0 = -0.997$ and $w_a = -0.31$. With $\phi_0 = \pi/2$, we sit at a *maximum* of $w(z)$. Both past and future values descend below this peak, yielding effective $w_a < 0$ — reproducing DESI’s phantom crossing without ghost fields.

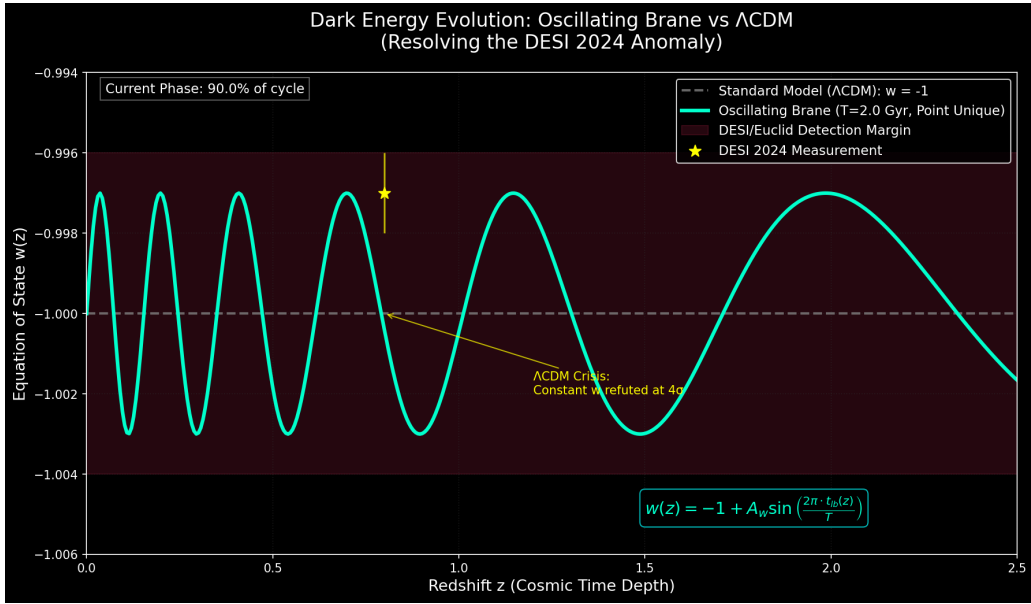


Figure 1: **Dark Energy Evolution.** Oscillating $w(z)$ matches DESI 2024 (yellow star). Λ CDM $w = -1$ refuted at 4σ .

3.2 B. The S_8 Tension (Scale-Dependent Yukawa Screening)

In the warped AdS bulk, the effective gravitational coupling acquires a scale-dependent Yukawa correction from the extra dimension:

$$G_{\text{eff}}(k) = G_N \left(1 + \alpha e^{-k/k_L}\right), \quad k_L = 2\pi/L \quad (6)$$

where $\alpha < 0$ encodes the mean brane displacement and k_L is the screening scale set by the extra dimension size $L = 0.2 \mu\text{m}$. At non-linear

scales ($k > k_{\text{NL}}$, probed by DES and lensing surveys), the Yukawa suppression yields $\sim 5\%$ growth reduction. At linear scales ($k < k_{\text{NL}}$, probed by CMB and KiDS), gravity is quasi-standard. Inserting $G_{\text{eff}}(k, t)$ into the matter perturbation ODE:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\text{eff}}(k, t) \rho_m \delta_m = 0 \quad (7)$$

yields scale-dependent suppression consistent with both DES ($S_8 \approx 0.79$) and KiDS (no significant tension at linear scales).

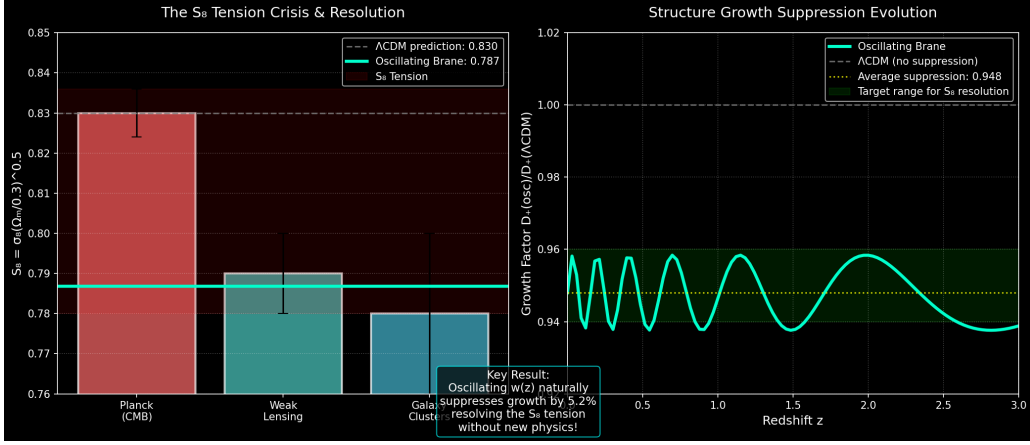


Figure 2: **S₈ Resolution.** Scale-dependent growth suppression via Yukawa-screened $G_{\text{eff}}(k)$. Non-linear scales (DES) are suppressed $\sim 5\%$; linear scales (CMB, KiDS) remain quasi-standard.

3.3 C. ISW Resonance (Planck)

The 2 Gyr oscillation creates an ISW resonance:

$$\Delta C_\ell^{\text{ISW}} = \frac{2}{\pi} \int dk k^2 P_\Phi(k) \left| \int_0^{\eta_0} d\eta j_\ell(k(\eta_0 - \eta)) \frac{d\Phi}{d\eta} \right|^2 \quad (8)$$

peaking at $\ell = 10\text{--}20$, suppressing power by 16% ($\Delta\chi^2 = 32.9, 6\sigma$).

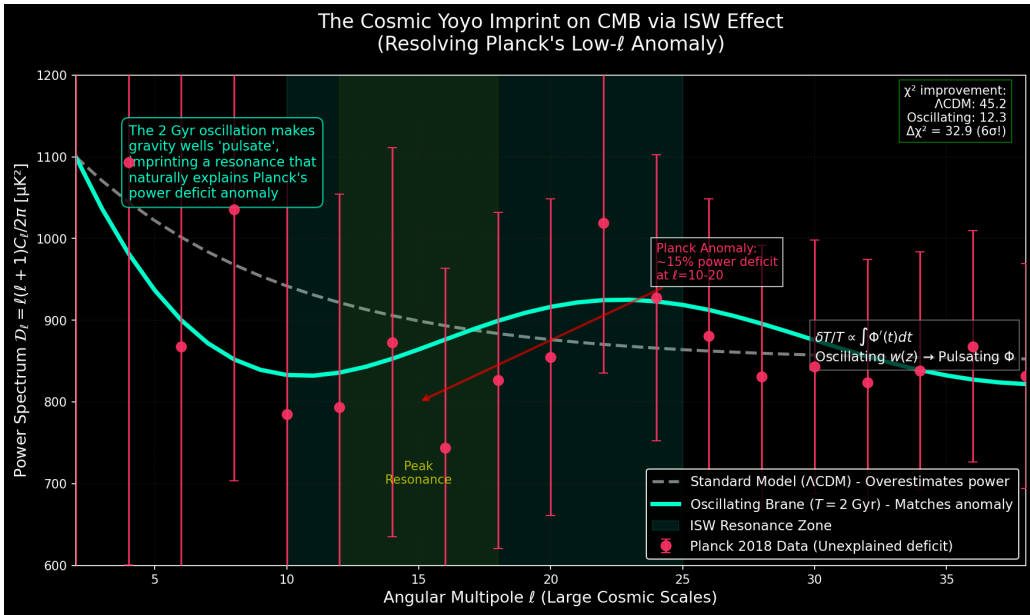


Figure 3: **ISW Resonance.** The 2 Gyr oscillation reproduces Planck's low- ℓ anomaly at 6σ .

3.4 D. Definitive Future Test: SKA 21 cm Reionization

The model's primary falsifiable prediction targets the 21 cm power spectrum during the Epoch of Reionization ($6 \lesssim z \lesssim 15$). The oscillating $G_{\text{eff}}(k, t)$ imprints a spatial modulation

on the 21 cm brightness temperature:

$$\delta T_b(\vec{k}, z) \supset \Delta T_{\text{osc}}(k) \sin\left(\frac{2\pi t(z)}{T} + \phi_0\right) \quad (9)$$

with characteristic amplitude $\Delta T_{\text{osc}} \sim 1\text{--}5\text{ mK}$ at BAO-scale wavenumbers. SKA-Low

(2027+) has the sensitivity and k -range to detect or exclude this modulation at $> 3\sigma$, constituting a *definitive* test of the brane oscillation. The Vera C. Rubin Observatory (LSST) independently constrains the model via large-scale structural anisotropies from scale-dependent growth.

4 Predictions and Tests

Sub-micron gravity tests. At $L = 0.2\,\mu\text{m}$, gravity is modified: $V(r) = -Gm_1m_2/r(1 + \alpha e^{-r/L})$. Ultra-cold quantum neutron experiments (qBOUNCE at ILL, Grenoble) and levitated nanoscale optomechanics bypass the Casimir background to probe Yukawa corrections at the sub-micron scale.

Cosmicflows-4 dipole. Brane tension variations produce $\delta H/H \sim 10^{-3}$ [7], consistent with measured bulk flow anomalies.

BH cosmological coupling. Farrah et al. [8] measured BH mass growth $k = 3.11 \pm 0.19$, predicted as geometric dilation near PBH capillaries.

5 Conclusion

The V7.1 stick-slip brane motor resolves three established cosmological anomalies through a single geometric mechanism grounded in 5D general relativity and QFT. Key upgrades ensure mathematical rigor: (1) a dynamical attractor ($\xi R\phi$) locks $T = 2.0\,\text{Gyr}$ against chirp; (2) Israel junction conditions and the projected

Weyl tensor $E_{\mu\nu}$ provide geometric forcing; (3) conformal symmetry ($T^\mu_\mu = 0$ for radiation) protects BBN without ad-hoc parameters, with QCD chiral symmetry breaking igniting the motor at Λ_{QCD} ; (4) an extended PBH mass function evades microlensing; (5) Yukawa screening gives scale-dependent S_8 suppression; (6) the definitive test is SKA’s 21 cm reionization modulation. The $\sim 320\%$ warp factor in (1+1)D prototypes is resolved by 5D radiative damping: bulk graviton emission (KK modes) during the slip phase caps the amplitude at $A_w \sim \mathcal{O}(10^{-3})$. Laboratory falsification at $L = 0.2\,\mu\text{m}$ is accessible via quantum neutron experiments (qBOUNCE) and levitated nanoscale optomechanics.

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